FIREWALL ASSIGNMENT

Task 1: Install and enable UFW

Run in the terminal:

```
sudo apt update
sudo apt install ufw -y
```

Check installation:

```
ufw --version
```

Task 2: Allow SSH and block other incoming connections

Add the SSH rule before enabling the firewall to reduce the risk of losing remote access.

```
sudo ufw allow in 22/tcp
sudo ufw default deny incoming
```

Set the outgoing default policy to deny:

```
sudo ufw default deny outgoing
```

This blocks other incoming connections and, by default, all outgoing connections unless explicitly allowed.

program for managing a netfilter firewall

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Enable UFW:

sudo ufw enable

Verify:

sudo ufw status verbose

Task 3: Allow outgoing HTTP and HTTPS only

Run:

```
sudo ufw allow out 80/tcp
sudo ufw allow out 443/tcp
```

Check the rules:

```
sudo ufw status numbered
```

The intended configuration is:

- Incoming TCP port 22: allowed.
- Other incoming connections: denied by default.

- Outgoing TCP ports 80 and 443: allowed.
- Other outgoing connections: denied by default.

Note: This restrictive configuration also blocks DNS requests, package downloads over other ports, and other outbound services. You may need additional rules for those services in a real environment.

Task 4: Save all current UFW rules

Run:

sudo ufw status verbose | tee firewall_rules.txt

Verify the saved file:

cat firewall_rules.txt

This creates firewall_rules.txt in your current directory and displays the saved output.

Task 5: Short explanation for your assignment

The UFW firewall allows incoming SSH connections on port 22 and blocks other incoming connections. Outgoing HTTP and HTTPS traffic on ports 80 and 443 is permitted, while other outbound connections are blocked. A local web server can run, but external users cannot access it unless its listening port is explicitly allowed. The development server may also be unable to connect to external APIs or databases because other outbound traffic is blocked.